

CodeOrbit Tech – Software Development Internship
Task 1: Requirement Analysis & SRS Document

SOFTWARE REQUIREMENTS SPECIFICATION

for

Task Manager Application

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Faculty / Internship Guide:	N/A
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2. Introduction

The Task Manager Application is a small software application designed to help users create, organize, prioritize, and track their tasks. This document provides a simple and clear Software Requirements Specification (SRS) for the application as part of Task 1 of the internship.

3. Purpose

The purpose of the Task Manager Application is to provide users with a simple way to manage their daily tasks. Users can create tasks, assign priorities and due dates, update tasks, mark tasks as completed, and search or filter their tasks.

4. Scope

- **In-Scope Features:** User registration and login, adding new tasks, viewing tasks in a list, editing task details, deleting tasks, marking tasks as completed, setting task priority and due dates, searching tasks by keyword, filtering tasks by status or priority, and logging out.
- **Out-of-Scope Features:** Multi-user team collaboration, third-party calendar synchronization, and automated external notifications.

5. Intended Users

- **Students and Interns:** To manage assignment deadlines, study schedules, and project tasks.
- **General Users:** To keep track of daily personal activities and to-do lists.
- **Evaluators and Mentors:** To review the software requirements and system features.

6. Overall Description

6.1 Product Perspective

The Task Manager Application is a simple software system that provides users with features to create, view, update, delete, and manage their tasks.

6.2 Product Functions

- **User Authentication:** Register, Login, and Logout.
- **Task Management:** Add task, view tasks, edit task, and delete task.
- **Task Organization:** Mark as completed, set priority (Low, Medium, High), and set due date.
- **Task Search & Filter:** Search by keyword and filter by status or priority.

6.3 User Characteristics

Users need basic familiarity with using a computer or mobile web browser. No technical or programming knowledge is required.

6.4 Assumptions and Dependencies

Assumptions: Users have access to a device with a web browser. Each registered user uses a unique email address.

Dependencies: The application requires an active web browser and basic data storage.

7. Functional Requirements

Req ID	Requirement Name	Description
FR-01	User Registration	The system shall allow new users to register by providing a name, email address, and password.
FR-02	User Login	The system shall allow registered users to log in with their email and password.
FR-03	User Logout	The system shall allow logged-in users to log out safely from the application.
FR-04	Add Task	The system shall allow users to create a new task with a title, description, priority, and due date.
FR-05	View Tasks	The system shall display all tasks created by the user in an organized list.
FR-06	Edit Task	The system shall allow users to update the details of an existing task.
FR-07	Delete Task	The system shall allow users to delete a task after confirming the action.
FR-08	Complete Task	The system shall allow users to mark a task as completed or unmark a completed task.
FR-09	Set Task Priority	The system shall allow users to set task priority as Low, Medium, or High.
FR-10	Set Due Date	The system shall allow users to set and update a due date for each task.
FR-11	Search & Filter Tasks	The system shall allow users to search tasks by keyword and filter tasks by status or priority.

8. Non-Functional Requirements

Req ID	Category	Description
NFR-01	Performance	The system should respond to user actions and load pages quickly without noticeable delay.
NFR-02	Security	The system shall ensure that user passwords and task data are protected from unauthorized access.
NFR-03	Usability	The user interface shall be clean, simple, and easy to navigate for users without special training.
NFR-04	Reliability	The system should perform consistently and prevent data loss during regular usage.
NFR-05	Availability	The application should be accessible whenever the user needs to manage tasks.
NFR-06	Maintainability	The system should be structured clearly so that future updates and feature additions are straightforward.
NFR-07	Scalability	The system should be able to handle an increasing number of tasks and users smoothly.

9. System Requirements

9.1 Hardware Requirements

- Computer / Laptop
- Minimum 4 GB RAM
- Keyboard and mouse

9.2 Software Requirements

- **Operating System:** Windows, Linux, or macOS
- **Web Browser:** Modern web browser such as Google Chrome, Mozilla Firefox, or Microsoft Edge

10. Use Cases

UC-01: Login

Actor	Registered User
Description	Allows a registered user to log into the application.
Preconditions	The user must have a registered account.
Main Flow	1. User opens the login page. 2. User enters registered email and password. 3. User clicks the Login button. 4. System validates the credentials. 5. System displays the task dashboard.
Alt/Exception	If credentials are invalid, the system displays an error message asking the user to try again.
Postconditions	User is logged in and can view the task dashboard.

UC-02: Add Task

Actor	User
Description	Allows the user to create and save a new task.
Preconditions	The user must be logged in.
Main Flow	1. User clicks the Add Task button. 2. User enters task title and description. 3. User selects priority and due date. 4. User clicks Save. 5. System saves the task and shows it on the dashboard.
Alt/Exception	If the title is left empty, the system asks the user to enter a title.
Postconditions	A new task is created and listed on the dashboard.

UC-03: View Tasks

Actor	User
Description	Allows the user to see all their existing tasks.
Preconditions	The user must be logged in.
Main Flow	1. User opens the dashboard. 2. System loads and displays all tasks with their title, priority, due date, and status.
Alt/Exception	If there are no tasks, the system displays a message stating that no tasks are available.
Postconditions	User can view the complete list of tasks.

UC-04: Edit Task

Actor	User
Description	Allows the user to modify an existing task.
Preconditions	The user must be logged in and the task must exist.
Main Flow	1. User selects the task to edit and clicks Edit. 2. System displays the task details in an editable form. 3. User modifies the necessary fields. 4. User clicks Update. 5. System saves the updated task details.
Alt/Exception	If the user cancels, no changes are saved.
Postconditions	The task details are updated in the application.

UC-05: Delete Task

Actor	User
Description	Allows the user to remove a task.
Preconditions	The user must be logged in and the task must exist.
Main Flow	1. User clicks the Delete button for a task. 2. System asks for confirmation. 3. User confirms the deletion. 4. System removes the task from the list.
Alt/Exception	If the user cancels the confirmation, the task is not deleted.
Postconditions	The selected task is permanently deleted.

UC-06: Complete Task

Actor	User
Description	Allows the user to mark a task as completed.
Preconditions	The user must be logged in and the task must exist.
Main Flow	1. User clicks the checkbox or Complete button on a task. 2. System updates the task status to Completed. 3. The task is displayed as completed on screen.
Alt/Exception	If the user unchecks the completed task, the system changes its status back to Pending.
Postconditions	The task status is updated.

UC-07: Set Priority & Due Date

Actor	User
Description	Allows the user to assign priority and deadline to a task.
Preconditions	The user must be logged in and creating or editing a task.
Main Flow	1. User chooses priority level (Low, Medium, or High). 2. User selects a due date from the date selector. 3. User saves the task. 4. System records the priority and due date.
Alt/Exception	If an invalid date is selected, the system prompts the user to select a valid date.
Postconditions	Task priority and due date are saved.

UC-08: Search / Filter Tasks

Actor	User
Description	Allows the user to find specific tasks or filter by category.
Preconditions	The user must be logged in and viewing the dashboard.
Main Flow	1. User types a keyword in the search bar OR selects a filter (All, Pending, Completed, Priority). 2. System filters and shows only matching tasks.
Alt/Exception	If no tasks match, the system displays 'No tasks found'.
Postconditions	Filtered tasks are displayed on the dashboard.

UC-09: Logout

Actor	User
Description	Allows the user to safely exit the application session.
Preconditions	The user must be logged in.
Main Flow	1. User clicks the Logout button. 2. System closes the active session. 3. System redirects the user to the login screen.
Alt/Exception	None.
Postconditions	User is logged out successfully.

11. User Stories

- **US-01 (Account Access):** "As a user, I want to register and log in, so that I can access my private task list."
- **US-02 (Task Creation):** "As a user, I want to add a new task with priority and due date, so that I can keep track of what I need to do."
- **US-03 (Viewing Tasks):** "As a user, I want to view all my tasks in one place, so that I can see my upcoming workload clearly."
- **US-04 (Task Modification):** "As a user, I want to edit my existing tasks, so that I can update details when plans change."
- **US-05 (Task Deletion):** "As a user, I want to delete unwanted tasks, so that my task list stays clean and organized."
- **US-06 (Mark Completed):** "As a user, I want to mark tasks as completed, so that I can track my daily progress."
- **US-07 (Search & Filter):** "As a user, I want to search and filter my tasks, so that I can quickly find urgent or pending tasks."
- **US-08 (Logout):** "As a user, I want to log out of the application, so that my account remains secure."

12. UML Use Case Diagram

The UML Use Case Diagram below shows the primary actor (User), the system boundary, and all functional use cases:

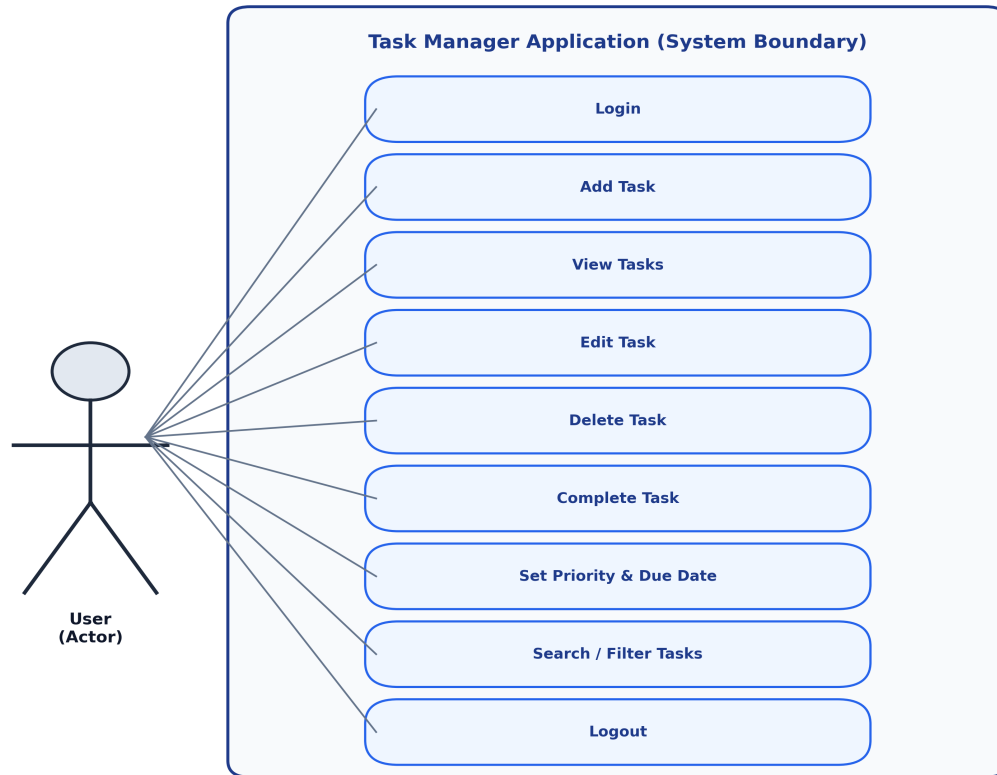


Figure 12.1: UML Use Case Diagram for Task Manager Application

13. Requirements Summary

The summary tables below present the complete set of Functional and Non-Functional Requirements:

13.1 Functional Requirements Summary

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NFR-06	Maintainability	The system should be structured clearly so that future updates and feature additions are straightforward.
NFR-07	Scalability	The system should be able to handle an increasing number of tasks and users smoothly.

14. Conclusion

The proposed Task Manager Application provides a simple and effective software solution for managing daily tasks and improving personal productivity. By covering all core requirements—such as user login, creating tasks, editing tasks, deleting tasks, setting priorities, assigning due dates, marking tasks as completed, and searching/filtering tasks—the system makes task organization simple and straightforward.

This Software Requirements Specification document clearly defines the functional requirements, non-functional requirements, use cases, and system requirements. It provides a complete and well-organized foundation for

Task 1 of the CodeOrbit Tech Software Development Internship.